Bismillahir Rahmanir Rahim

International Islamic University Chittagong

Department of Computer Science and Engineering

Mid-term Examination, Spring-2022	Program: BSc in CSE
Course Title: Electrical Drives and Instrumentation	Course Code: EEE-2421
Time: 1hour 30 minutes	Full Marks: 30

1(a)	State Fleming Left Hand Law. Why Motor sizing is important?	U	CO2	2
1(b)	Describe the different types of motor duty class with proper timing diagram.	R	CO2	4
1(c)	State the four basic principle of magnetic fields applied on electrical machines and justify the statement - 'Magnetic fields are the fundamental mechanism by which energy is converted from one form to another in transformer'.	An	CO1	4
2(a)	"A DC machine can work as both the generator and the motor" Justify the statement.	U	CO2	2
2(b)	The load torque profile shown in Fig. 2(b) is measured for a constant speed load. Choose an appropriately sized motor to drive this load. (Assume a service factor of 25%). You do not need to consider starting performance	An	CO1	3

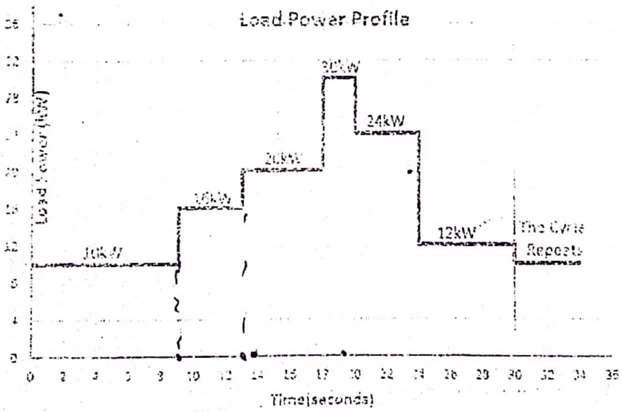


Fig. 2(b)

2(c)	"A DC generator can be made from an AC generator" Describe the operation of DC generator with neat sketch and also justify the statement.	An	CO1	5

OR

2(c)	Draw the equivalent circuit of a transformer with proper notation. You build a 220V/9V transformer by yourself and want to draw an equivalent circuit of your transformer. How do you get values of all the parameters of the equivalent circuit? Write in detail.	An	CO1	5

CamScanner

3(a) "We cannot think of our modern power system without transformer." Justify the statement. U CO2 2

3(b) Briefly explain the different types of losses in a transformer. Draw a three phase delta-wye connection. R CO2 3

3(c) The open circuit test and short circuit test were performed on the primary side of a 20-KVA, 8000/240-V, 50-Hz transformer and the following data were taken: An CO1 5

Open circuit test (on primary)	Short circuit test (on primary)
$V_{OC} = 6000\ V$	$V_{SC} = 367\ V$
$I_{OC} =$ [illegible]4 A	$I_{SC} = 2.5\ A$
$P_{OC} =$ 3[illegible] W	$P_{SC} = 180\ W$

Find the impedances of the approximate equivalent circuit referred to the primary side and sketch the circuit.

OR

3(c) A 100-KVA, 2000/440-V transformer has $R_1 = 3\ \Omega, R_2 = 0.07\ \Omega$. The values of reactances are $X_1 = 5.2\ \Omega, X_2 = 0.075\ \Omega$. An CO1 5

i) Calculate equivalent impedance referred to primary.

ii) Calculate equivalent impedance referred to secondary.

R_1 R_2'

International Islamic University Chittagong
Department of Computer Science and Engineering (CSE)

B.Sc. Engineering in CSE

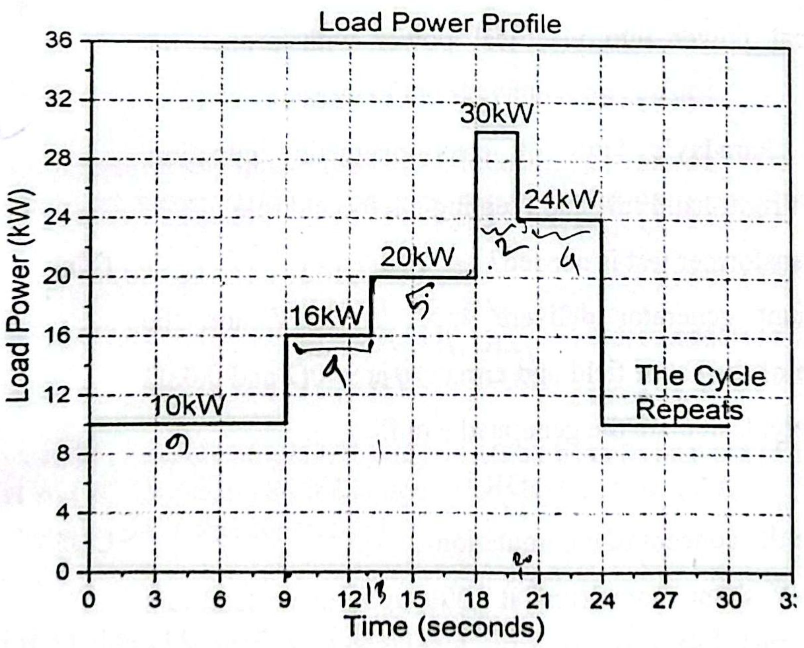
Midterm Examination, Spring 2025

Course Code: EEE-2421 **Course Title:** Electrical Drives and Instrumentation

Time: 1.5 hours **Full Marks:** 30

Answer all the questions. The figures in the right-hand margin indicate full marks
Course Outcome (COs) and Bloom's Level (BL) are mentioned in additional Columns

1(a)	i. List the different types of motor duty classes and explain any two of them with timing diagram. ii. The magnetic flux linked with a coil having 250 turns is changed from 1.4 Wb to 2 Wb in 0.45 seconds. Calculate the induced e.m.f in the coil. 333.33	CO1	Ap	5
1(b)	The following load power profile was measured for a constant speed load. Choose an appropriately sized motor to drive this Load. Assuming a service factor of 30%. You do not need to consider starting performance.	CO3	Ap	5



79.25%

2 (a)	i. What is transformer? Draw the equivalent circuit of transformer with proper notation. ii. Describe the core losses in a transformer.	CO1	R	5

2 (b)	i. Classify the transformer test. ii. In no-load test of single-phase transformer, the following test data were obtained: Primary voltage: 230V; Secondary voltage: 115V Power Input: 40W; Primary Current: 0.5A Find the following: (i) the turns ratio 2 (ii) the magnetizing component of no-load current 0.4688 (iii) its working component 0.1739 (iv) the iron loss $\boxed{R=0.6}$	CO3	An	1+4
	OR			
2 (b)	i. Distinguish between the transformer test. ii. A 5 kVA 200/1000 V, 50 Hz, single-phase transformer gave the following test results: O.C. Test (L.V. Side): 200 V, 1.2 A, 90 W S.C. Test (H.V. Side): 50 V, 5A, 110 W Calculate the parameters of the equivalent circuit referred to the L.V. side.	CO1	An	1+4
3 (a)	i. Describe the operation of DC machine that converts mechanical power into electrical power with proper net sketch. ii. State Faraday's laws of electromagnetic induction, Fleming's right-hand rule and Fleming's left-hand rule.	CO1 V=IaRa	R	5
3 (b)	i. Why transformer test is needed? ii. A shunt generator delivers 455A at 230V and the resistance of the shunt field and armature are 40Ω and 0.03Ω respectively. Calculate the generated e.m.f.	CO3 V−IaRa 250−(	An	1+4
	OR			
3 (b)	i. Analyze the concept of commutation. ii. A 255-V shunt motor runs at 1000 r.p.m at no-load and takes 9A. The total armature and shunt field resistances are respectively 0.2Ω and 255Ω. Calculate the speed when loaded and taking 50A. Assume the flux to be constant.	CO3	An	1+4

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE

Mid-term Exam, Spring 2023

Course Code: EEE-2421

Course Title: **Electrical Drives and Instrumentation**

Time: 1 hours 30 minutes

Full Marks: 30

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

1	a)	Analyze Linear DC Machine as a Motor.	CO1	E	4
1	b)	Explain hysteresis loss and eddy current loss in a ferromagnetic core ?	CO1	U	3
1	c)	List the reasons for which it is important to size a motor? Discuss How to choose the right motor briefly?	CO1	Ap	3
2	a)	Compare the characteristics of an Ideal and a practical transformer. Briefly Discuss the steps to eliminate the losses of a transformer?	CO2	U	3
2	b)	A 2200/200V transformer draws a no-load current of 0.6A and absorbs 400 watts. Find the magnetizing and iron loss currents.	CO2	An	3
2	c)	A 5 kVA 200/1000 V, 50 Hz, single-phase transformer gave the following test results : O.C. Test (L.V. Side) : 200 V, 1.2 A, 90 W S.C. Test (H.V. Side) : 50 V, 5A, 110 W Calculate the parameters of the equivalent circuit referred to the L.V. side.	CO2	An	4
		Or			
2	a)	Discuss the working principle of a transformer and draw a vector diagram of an ideal transformer?	CO1	U	3
2	b)	A 50-kVA, 4,400/220-V transformer has $R_1 = 3.45\ \Omega$, $R_2 = 0.009\ \Omega$. The values of reactances are $X1 = 5.2\ \Omega$ and $X2 = 0.015\ \Omega$. Calculate for the transformer (i) equivalent resistance as referred to primary (ii) equivalent resistance as referred to secondary (iii) equivalent reactance as referred to both primary and secondary (iv) equivalent impedance as referred to both primary and secondary	CO2	An	4

CamScanner

CamScanner

2	c)	In no-load test of a transformer, the following test data were obtained: Primary Voltage: 220V; Secondary voltage: 110V Power input: 30W; Primary Current: 0.5A Find the following: i)The turns ratio ii) Magnetising component of no-load current iii)Its working component iv) iron loss.	CO2	An	3
3	a)	Write a short note on the basic principle of a single loop DC generator.	CO1	U	2
3	b)	Briefly Discuss on armature reaction which occurs in electrical machines with necessary diagrams.	CO1	U	4
3	c)	A 250-V shunt motor runs at 1000 r.p.m. at no-load and takes 8A. The total armature and shunt field resistances are respectively 0.2 Ω and 250 Ω. Calculate the speed when loaded and take 50 A. Assume the flux to be constant.	CO2	An	4

CamScanner
CamScanner